

No. of Functional Features Included in the Solution

Date	28 September 2026
Team ID	6ab4d2876603573ccfb56348
Project Name	AI WEATHERWISE
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Secure user signup and login with hashed passwords and JWT token issuance.	Authroutes / authcontroller	Done	High
2	Real-Time Weather Retrieval	Fetches real-time temperature, condition, humidity, and wind speed for a target location.	Weatherroutes / weathercontroller	Done	High
3	AI Summary & Recommendations	Generates automated weather summaries and actionable daily recommendations using AI.	Airoutes / aicontroller	Done	High
4	Saved Locations Management	Allows users to add, retrieve, and delete favorite geographical locations.	Locationroutes / locationcontroller	Done	Medium
5	JWT Authentication Middleware	Protects restricted routes by validating authorization headers and JWT bearer tokens.	Authmiddleware	Done	Medium
6	MongoDB Persistence Layer	Defines structured schemas (User, Location) and handles database connection error handling.	Config/db.js / models	Done	Medium
7	Input Validation & Error Handling	Validates incoming payloads and returns structured JSON error responses.	App.js / controllers	Done	Low
8	Interactive Map Visualization	Interactive visual map rendering weather metrics across geographical layers.	Frontend / maps component	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Search bar for location input, weather info cards, and error toast alerts.
2	Backend / Logic	2	Express routing, controller logic, and AI recommendation generation service.
3	Database / Storage	2	Mongoose ODM schemas for Users and Saved Locations connected to MongoDB.
4	API / Integration	2	External Weather API integration and AI weather summary service integration. Password hashing (bcrypt) and JWT middleware route protection.
5	Security / Authentication	1	Password hashing (bcrypt) and JWT middleware route protection.